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Innovation in Designing Drilling Bits: Integrating Simulations and Bit Cutter Re-Design to Improve Bit Performance

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Abstract

In deep, abrasive, highly interbedded reservoirs with compressive strengths ranging from 15,000 – 30,000 psi, drilling long drain holes sections with current drilling bit designs proved ineffective. Bit interactions with the formation resulted in high erratic torque, requiring reduction of drilling parameters to a conservative level. This resulted in low rate of penetration (ROP) and ultimately impacting drilling performance. This necessitated the re-evaluation of the bit-formation compatibility to derive possible bit design augmentations to improve ROP.

A detailed analysis was performed on previous bit runs with emphasis on the bits design, cutter technology and dull gradings. This revealed that the bit design could be optimized for drilling longer laterals across these formations while improving ROP performance and bit life. New bit design models based on innovative cutter geometry and bit structural design were run through intensive simulation programs to test their durability and torque levels during drilling. The simulations showed improved results from existing runs, and new trial test bits were manufactured from these designs. These new bits were ultimately deployed in selected wells for evaluation.

The trial runs with these newly engineered bits yielded a 70% improvement in drilling ROP across the formations in question. The bits also showed better torque stability throughout the drilling run and drilled longer footage compared to previous runs. The bit life also improved, as these bits drilled over 5000 ft of hole section in a single run, unlike in previous wells that required multiple bit-trips to drill the same footage. Previous wells also required additional reaming trips prior to pulling out of hole due to irregular wellbore geometry. However, on the trial runs with the newly designed bit, the wellbore geometry was considerably better and did not require reaming trips. This was mainly due to the reduced erratic torque and improved bit stability. Post-run bit dull grade analysis showed increased frontal impact resistance, improved durability and higher resistance to wear. Open hole logs were performed without encountering any obstructions or tight spots and subsequent completions were run across the open hole without issues. Ultimately, this re-engineered bit solution led to a 5-day average reduction in well delivery time per well when fully implemented. This resulted in substantial cost savings and improved well delivery performance across the field.

This paper will discuss the entire bit re-design and implementation process, from initial analysis of previous bit runs to the actual bit re-design and manufacturing, the successful trial runs, and review of the trial run results. The success achieved with this bit re-design process will open doors for further improvements in bit design and ultimately result in improve drilling performance and drilling cost reduction.

Introduction

Recently, there has been a very strong push by major stakeholders in the oil and gas industry to accelerate well delivery by adopting significant advancements in drilling operations and enhancing performance. To ensure that drilling operations remain profitable and sustainable, the significance of enhanced well delivery cannot be overstated.

To enhance production from deep gas reserves, the strategy has shifted from drilling multiple vertical wells to drilling fewer wells with long laterals to improve reservoir drainage efficiency. The importance of designing a fit-for-purpose drilling system for optimum drilling performance across these abrasive formations can therefore not be over-emphasized.

The well is drilled laterally through the gas-bearing abrasive and interbedded carbonate reservoir formation. Given the fact that the formations are highly abrasive and interbedded, it is crucial to develop a BHA that tackles all the recognized and expected drilling challenges.

A number of initiatives were previously launched to address drilling issues, including persistent string stalling, string buckling, excessive stick-slip, significant shocks and vibrations, frequent bit failures, and downhole equipment malfunctions. Various bit designs were tried, and there were alternations between Motor BHA, Motorized-RSS BHA and RSS stand-alone BHAs, but these efforts did not lead to a notable decrease in drilling difficulties. Ultimately, a decision was made to adopt more conservative, reduced, or safer drilling parameters. This approach helped lower the occurrence of tool failures and the need for unplanned trips to change the BHA. However, this solution proved unsustainable since the downtime caused by unexpected tool failures was less than the unseen time lost due to a reduced rate of penetration (ROP) or the increased cost per foot.

Offset Wells Analysis

The offset wells analysis provided the clear understanding of the following:

- Drilling challenges encountered in previous wells drilled on the project which includes:
 - i. Wellbore stability issues
 - ii. Stuck pipe mechanisms
 - iii. Downhole equipment failures due to high stick/slip leading to unplanned trips
 - iv. Downhole equipment failures due to high shock and vibration issues
 - v. Hole cleaning limitation due to high circulating pressures
- Formation characteristics (rock properties such as formation abrasiveness and hardness)
- Review of the different mud systems implemented historically.

It was also necessary to carefully examine the numerous drilling systems previously implemented on the field to see how the impact of their differences can be accumulated into the BHA re-design process.

A close analysis of over 300 horizontal wells drilled on the project revealed that 30% of total well time was consumed in the 5-7/8" hole section. Most of this time was consumed by slow drilling operation due to inability to apply the optimum drilling parameters in order to avoid unplanned trips due to downhole equipment failures. On wells where optimum parameters were applied, there were several unplanned trips for bit or BHA change.

After reviewing the results of the offset wells analysis, it became clear that in order to address the results of the offset wells analysis, an engineered drilling systems solution had to be proposed that helps achieve the following:

- Ability to drill through highly abrasive and interbedded formation.
- Must be stable to avoid or reduce stick-slip, shocks and vibration.
- Optimize torque and drag.
- Reduction in flat time.

Formation Analysis: Lithology and Compressive Strength Plot

A critical analysis of the formation shows that compressive strength ranged from 15-30 Ksi and could exceed 30 ksi. The formation is also highly abrasive. To optimize performance, the right drill bit has to be selected, coupled with the right drive system and the right mud system. A complete review of the drilling system had to be performed in order to identify what will be done differently. The plot in [fig. 1](#) shows the formation analysis result from a well drilled on the field.

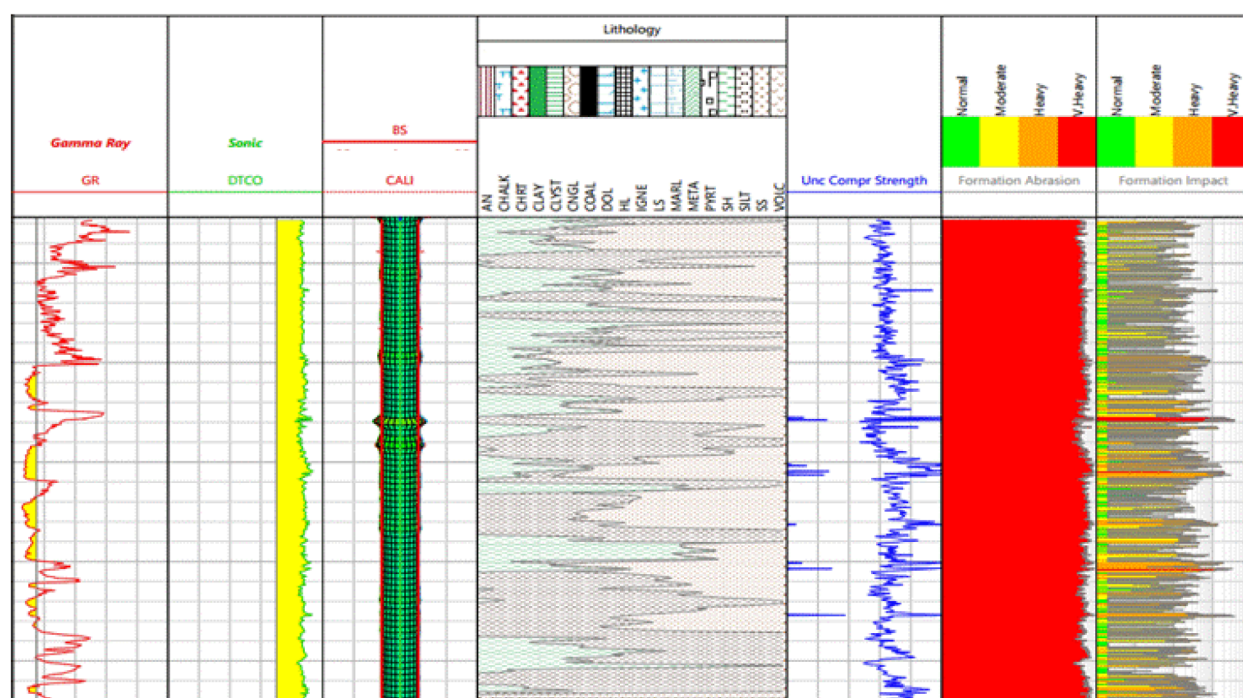


Figure 1—Rock Strength Analysis Software Result from one well in the field (Efe et al 2022)

Bit and drill string design

In the course of searching for solution to the numerous drilling challenges, different BHA designs were implemented. Depending on the length of the 5-7/8" lateral to be drilled, the typical BHA was designed with 4,000 – 6,000 ft of 4" drill pipes and then 5.5" drill pipes to surface. Drilling always resulted in high stick-slip, string vibrations, MWD failures, string buckling and constant damage and replacement of top drive bolts. The initial drilling system implemented was motor BHA in order to reduce the surface RPM requirement. This led to reduction in string vibration and TDS bolts damages and very marginal improvement in the overall drilling performance (ROP). A few tries were made with the motorized RSS BHA which also resulted in unplanned trips for bit change and downhole equipment failure. Switched

to stand-alone BHA which did not result in significant improvement in drilling performance, however equipment failure reduced slightly due to application of lower RPM at the bit. Due to inability to apply maximum WOB irrespective of the BHA type, it became necessary to review the bit design. Some of the other challenges or problems encountered historically included:

- Mechanical cavings occurrences due to disturbance by drill string.
- Very poor hole condition leading to last-minute changes in completion design from Open Hole Multi-Stage Frac completion to cemented liner completion.
- String twist-offs due to high levels of shocks and vibrations.
- Measurement-while drilling (MWD) or Logging-while-drilling (LWD) tools failure leading to unplanned trips.
- Bit cutting structure damages, resulting in very low ROP and unplanned trips.

After reviewing all the numerous BHA configurations implemented on the project, quickly identified that an engineering solution was required to further optimize the drill bit design. Performed finite-element-analysis for the different BHAs, found the BHAs to be relatively stable, however, the BHA bouncing, sting vibrations and stick-slip was mostly generated from the interaction between the bit and the formation.

As indicated earlier, the conventional drill PDC bit design, though delivering fairly average performance, did not address all the drilling risks during the design process. There was a lot of room for improvement.

Finite Element Analysis (FEA) and BHA modelling was carried out in a bid to generate a fit-for-purpose 5-7/8" Bit capable of withstanding drilling environment challenges while ensuring bit cutting structure durability. The model recommended a bit with the following changes:

- Increase in the number of gauge cutters
- Replace some of the conventional bit cutters with revolutionized cutters with thicker diamond structure for improved durability
- Optimize the cutter spatial placement on the Bit matrix
- Reduce the length of cutters from 16mm to 13 mm.
- Reduce the length of the bit's neck (distance between the cutting structure and the bit's pin end)
- Incorporate bit-gage protection diamond inserts to help minimize the rate of wear of gauge/face cutters.

This engineered bit design was much different from the previous conventional bit design. The new bit proved to be more stable, more durable, more resilient and delivered better directional control. The optimized bit design also resulted in improved BHA stability. This meant better transmission of applied parameters to the bit which resulted in higher drilling ROP. The new bit was also designed with higher WOB window, which meant a bigger operating envelope. Many runs have been performed with the improved bit and the finding is that irrespective of the drive mechanism applied (RSS or Motorized-RSS), the results are consistent and highly improved.

The newly improved drill bit has the following additional advantages:

- Flat time operations have been drastically reduced due to improved BHA stability which has in-turn reduced the number of downhole equipment failures
- Wells are delivered faster due to 70% ROP improvement well-on-well.
- Reduced resources consumption as back-up bits, MWD and LWD tools are now committed to other wells as they are no longer tied up for potential trips for BHA replacement.

- Improved HSE across the project due to reduce pipe handling. Historically each lateral required 2-3 BHA trips, these days, the section is drilled in one run (shoe-to-shoe).
- Elimination of wiper trips to prepare the lateral for completion run. The hole condition is much better now due to improved BHA stability which has greatly reduced formation disturbance and cavings.

It is important to note that when advanced drill string modelling was performed, results showed that conventional drill string was unable to fully transmit all the weight applied at the surface to the bit due to string vibrations, bit whirl, stick-slip and premature cutting structure damage, most of the transmitted weight gets dissipated and is not used productively to shear the formation to achieve desired drilling progress. Simulation showed that the optimized drill bit design gave better results due to increased operational window.

Drilling Dynamics and Parameters Optimization

Very high levels of stick slip were observed whenever the hole section was drilled with old conventional bit/drill string. This was the key reason why it was impossible to apply optimum drilling parameters. While drilling the hole section with RSS BHA and the conventional bit, WOB application and surface RPM had to be limited, resulting in a reduced average ROP which ultimately impacted drilling performance.

When the optimized bit was used to drill the same hole section in adjacent wells, stick-slip and vibration levels were drastically reduced. This allowed the application of optimized drilling parameters (RPM and WOB), leading to improved average ROP and reduction in drilling time.

It is important to note all the runs considered for the analysis utilized the same RSS BHA and mud system. The purpose of this was to see how the re-designed bit impacted the overall drilling performance while maintaining similar parameters.

Finite element Analysis

Based on design considerations from the software, several designs were simulated in an advanced FEA-based drilling dynamics simulation. The simulation modeled the interaction between the cutting structures and the formation based on the lab test result, and then presented the simulation result in a time-based manner. The input of the simulation consisted of several bit options, well trajectory, BHA details.

The graph in [fig. 2](#) shows lateral vibration along different BHAs when drilling through carbonate reservoir formation. Due to improved bit design implementation, observed reduce vibration along tool string. BHA #1 and BHA #2 were previous runs with the conventional bit design while BHA #3 is with optimized bit.

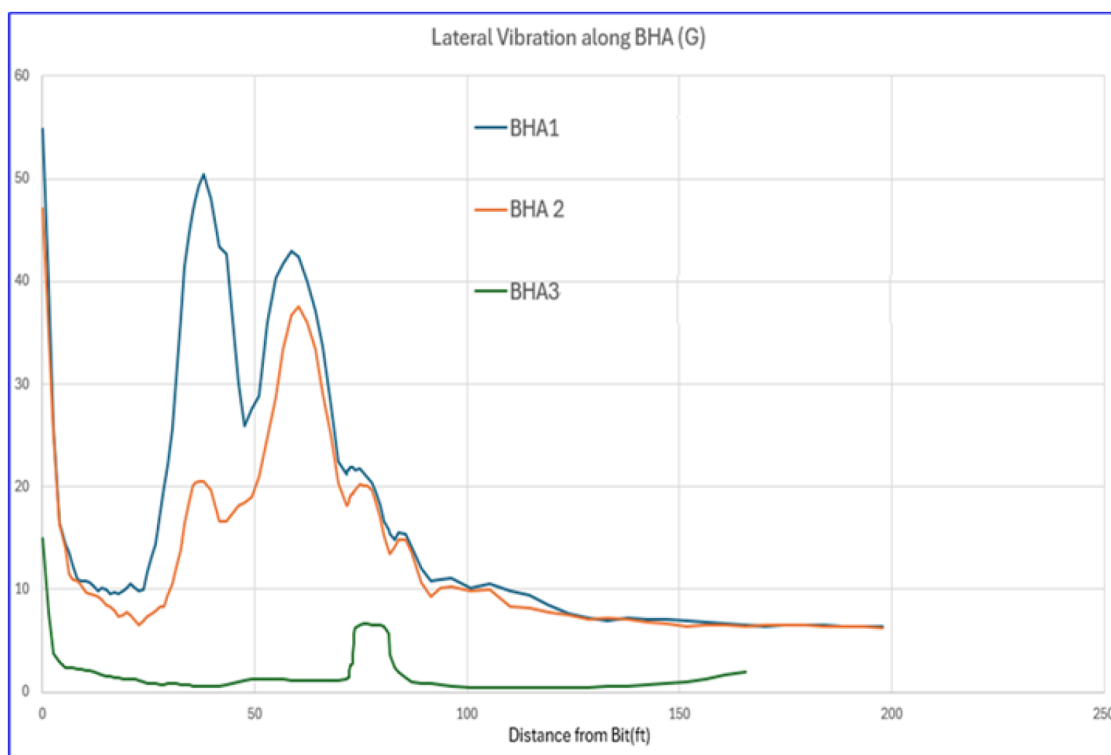


Figure 2—Lateral Vibration analysis for different bits compared to BHA with new bit design.

Bits Stability Comparison Analysis

Performed bit stability analysis of the conventional bit vs redesigned bit on the exact same RSS BHA. Due to improved interaction between the bit and the formation, observed better drilling dynamics report with the new bit compared to the conventional bit. There was reduced stick-slip and bit bouncing with the newly optimized bit design compared to the conventional bit. Refer to [fig. 3](#) for detailed simulation result.

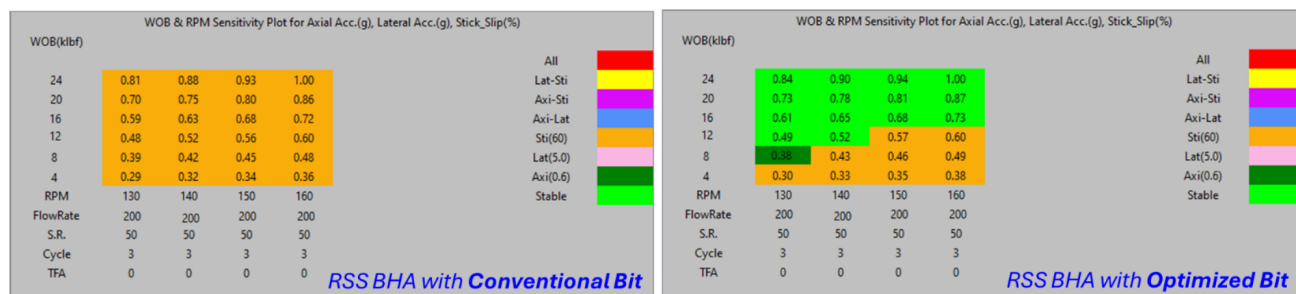


Figure 3—Bit stability analysis comparison.

General results from Solution Implementation

The optimized drill bit has led to great improvement in drilling efficiency by reducing stick-slip, vibrations, string stalling, and downhole equipment failure events while drilling the 5-7/8" hole section of the gas wells. This has resulted in increased rate of penetration (ROP), improved daily drilled footage, reduction in number of BHA trips due to downhole tool failure (bits and MWD tools) and reduction in overall well costs. For the operator, this has translated to faster well-to-production time.

The optimized drill bits have since been deployed on all subsequent wells drilled in this field. The results have been consistently good well-on-well. [Fig. 4](#) is a comparison of performance from conventional drill

bit design and the optimized drill bit design. Drilling performance has doubled. It is important to note that there was no additional cost incurred by drilling contractors in the process.

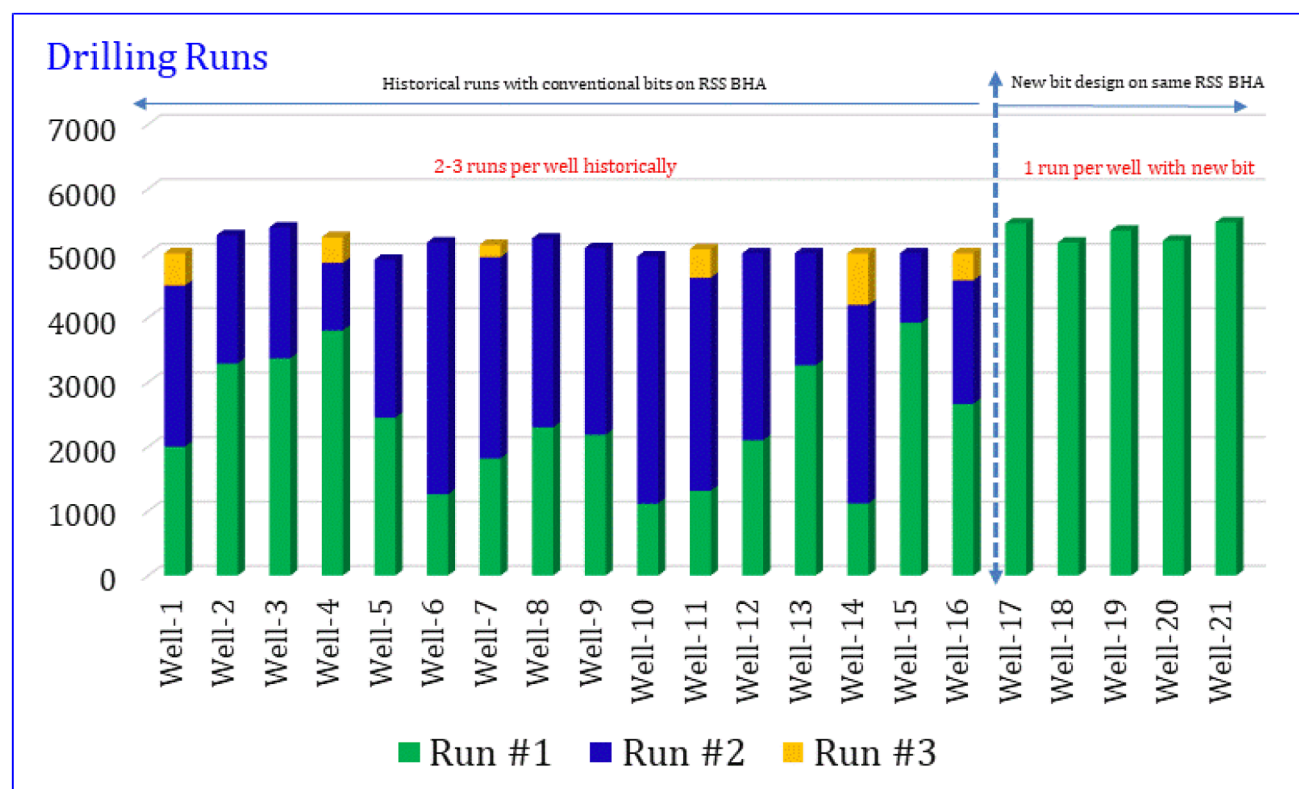


Figure 4—Impact of drill bit optimization on overall drilling performance.

Conclusion

The fit-for-purpose bit design has greatly improved operational efficiency, well costs and subsequently expedited wells delivery. This new solution has been implemented in all horizontal wells on this field with the same high level of performance, significant time savings and reduced HSE exposure.

In summary, the key wins observed after implementing this optimized solution include:

1. 50% reduction in drill string handling due to eliminated BHA trips. This translates to reduced HSE incidents.
2. The on-bottom and average drilling ROP have been doubled due to the optimized bit design which has enabled application of maximum recommended drilling parameters. The average savings per well due to improved ROP is 130 hours (5.42 days).
3. The new bit has shown increased frontal impact resistance, improved durability and higher resistance to wear due to the increased diamond in the cutters.
4. Reduction in the number of trips required to drill the long 5-7/8" lateral.
5. Better allocation and utilization of resources (drilling tools)
6. Subsequent wells showed lower/reduced drill string vibrations and stick/slip levels.
7. Wells drilled are now being delivered at least 5 days ahead of the plan due to innovative solution.
8. Reduction in mechanical cavings incidents due to improved BHA stability.
9. The quality of wellbore has improved after implementing the new drill bit.
10. Previously, prior to conducting dedicated open hole logging operation, there was a requirement to conduct a reaming trip to ensure good hole quality. The practice was to POOH drilling BHA (and lay

down the assorted jewelry), then RIH with the dumb iron rotary BHA to bottom, clear all tight spots and POOH for open hole wireline logging operation. On average, each reaming trip took 36 hours.

11. Historically, after performing extensive logging operation, a round trip was required to condition the drilling mud and clear any potential problem zone (intervals with big washouts or failures). On average, this trip took 24 hours. Due to the quality of the hole drilled these days, this condition trip is no longer required.
12. There has been drastic reduction in mechanical stuck pipe incidents while drilling this section. This is due to improved wellbore quality and reduced quantity of cavings generated from mechanical disturbance.
13. Since incorporating this improved drill bit/string design, the average number of trips for BHA change has been eliminated.
14. Each horizontal well drilled takes approximately 70 days. Typically, each rig is expected to drill 5 wells per annum. 6 days savings per well translates to 30 days saved per rig (or half a well equivalent saved per rig per annum).

Nomenclature

- ROP: Rate of Penetration.
- BHA: Bottom Hole Assembly.
- UCS: Unconfined Compressive Strength.
- WOB: Weight on Bit.
- POOH: Pull out of hole.
- RIH: Run in Hole.
- RPM: Revolutions Per Minute.
- Klbs: Kilo ponds of weight.
- Ft/hr: footage drilled per hour.
- PDC: Poly Crystalline Diamond Cutter.
- HSE: Health, Safety and Environment.
- ECD: Equivalent Circulating Density.
- HWDP: Heavy Weight Drill Pipe.
- DP: Drill Pipe.
- RSS: Rotary Steerable System.
- MWD Tool: Measurement While Drilling Tool.
- LWD Tool: Logging While Drilling Tool.

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